Solution

CADET _____ SECTION _____ TIME OF DEPARTURE _____

DEPARTMENT OF CHEMISTRY & LIFE SCIENCE

QUIZ 1 – CH402 2025-2026
6 Minutes, 24 Points
20 February 2026

TEXT: Peters, Timmerhaus, & West
SCOPE: Chapter 6

References Permitted: FE Reference Manual online.

INSTRUCTIONS

1. There are 4 problems on 1 page in this quiz (not including the cover page or supplements).
2. You will have 6 minutes for the quiz.
3. Do not mark this quiz until "begin work" is given.
4. Circle the correct answer to receive full credit.

(TOTAL WEIGHT: 24 POINTS)

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
A	6	
B	6	
C	6	
D	6	
TOTAL CUT		
GRADE	24	

Cadet: Solution

Problem:	Weight:
A	6

The delivered equipment cost for a solid-fluid processing plant is $250,000. On average, the fixed-capital investment for the plant is most nearly

(a) $610,000
(b) $760,000
(c) $860,000
(d) $1,100,000

Use Lang factors, FE page 257

$4.3 \times 250{,}000 = \$1{,}075{,}000 \cong \$1{,}100{,}000$ ANS

Problem:	Weight:
B	6

The delivered equipment cost for a solid-fluid processing plant is $250,000. On average, the working capital for the plant is most nearly

(a) $100,000
(b) $180,000
(c) $240,000
(d) $310,000

Use Lang factors

$(5.0-4.3) \cdot 250{,}000 = \$175{,}000 \cong \$180{,}000$ ANS

Problem:	Weight:
C	6

A six-foot stainless-steel bubble-cap tray cost $1,850 in 1999. Estimate the cost of a similar ten-foot tray in 2020. The chemical engineering price index factors are 435.5 and 764.7 for 1999 and 2020, respectively.

(a) $1,800
(b) $3,400
(c) $5,000
(d) $6,000

Use scaling factors, FE page 257-258

$\$1850 \cdot \left(\frac{10}{6}\right)^{1.2} \cdot \left(\frac{764.7}{435.5}\right) = \$5{,}996 \cong \$6000$ ANS

Problem:	Weight:
D	6

A fixed-tube-sheet shell-and-tube heat exchanger with an area of 120 m^2 cost $12,800 in 2006. What is the cost of a similar heat exchanger with an area of 300 m^2 in 2020? The chemical engineering price index factors are 548.0 and 650.1 for 2006 and 2020, respectively.

(a) $22,181
(b) $19,156
(c) $22,725
(d) $26,313

Use scaling factors

$\$12{,}800 \cdot \left(\frac{300}{120}\right)^{.44} \cdot \left(\frac{650.1}{548.0}\right) = \$22{,}725$ ANS